

Blaise Baptist

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CS student writing ROS 2 software for autonomous robots, seeking a Summer 2027 software internship

EDUCATION

Rose-Hulman Institute of Technology

Bachelor of Science in Computer Science; GPA 2.34

Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithm Analysis, Systems Programming, Object-Oriented Software Development, Digital Systems; *in progress*: Web Programming, Debugging

Terre Haute, IN

Expected May 2029

EXPERIENCE

Thornbots, Rose-Hulman ARC Robotics Team

Secretary; Software Developer, Sentry Robot

Sept 2025 – Present

Terre Haute, IN

- Built the ROS 2 localization stack for the autonomous Sentry robot, with SLAM, AMCL, and EKF backends that fuse wheel odometry with rf2o laser odometry
- Wrote the targeting stage of the CV pipeline, which tracks enemy robots from detected armor panels and decides where to aim and when to fire; co-wrote the depth step that turns 2-D detections into 3-D panels
- Wrote Gazebo test benches that measure localization drift against ground truth and score aiming and tracking
- Ported the full robot stack from ROS 2 Humble to Jazzy and Isaac ROS 4.6

Association Resource Center

Coding Intern

July – August 2025

Remote

- Built internal Rust tools for staff who manage many professional associations
- Wrote a scraper that pulls the tags out of client associations' articles, replacing a manual step for staff
- Worked with a team on the shared back end used across clients and on a front-end site in Rust, HTML, and CSS

PROJECTS

MediaManager | Python, Django, Rust, ffmpeg

Apr – May 2026

- Built a client-server system that splits AV1 encoding across 5 machines, saving 1.5 TB on a 6 TB media library
- The Django server syncs with Radarr and Sonarr and uses ffprobe to queue only files that need re-encoding
- It locks each job to one worker and requeues jobs from workers that stall
- The Rust client runs parallel jobs on the machine's hardware AV1 encoder, or on SVT-AV1 if it has none

Mobile Home Lab | Ubuntu, Docker, Tailscale, iptables

Aug 2025 – Aug 2026

- Set up a Protectli Vault as a portable router that shares an Ethernet or LTE uplink over its own Wi-Fi network
- Runs Pi-hole as DNS for the whole tailnet, with a second Pi-hole for failover, and works as a Tailscale exit node
- Bridged two Ethernet ports and the Wi-Fi into a single LAN so the router also works as a network switch
- Hosts Vaultwarden, Calibre-Web, and Netdata in Docker, plus Samba file shares and an rclone remote mount
- Rewired it to run for 21 hours over USB-C on a battery pack built from custom Li-ion cells

Newton Fractals | Rust

2024 – 2025

- Wrote a multithreaded renderer for Newton's-method fractals that splits work across all CPU cores with Rayon
- Has produced 10-gigapixel images and includes an interactive Iced viewer for zooming and panning
- Takes roots from a file or generates them at random, and computes the polynomial and its derivative itself

RS-232 Serial Port Reader | Rust, Iced

Sept 2024 – May 2025

- Built a desktop app with an Iced GUI that reads Arduino sensor data over a serial port
- Splits one port's stream into several live graphs and saves recordings as raw bytes or CSV

VEX Robotics | Python, Fusion 360

Aug 2021 – May 2025

- Designed competition robots in Fusion 360 and wrote a PID controller in Python for autonomous driving
- The team qualified for the state championship three times and the VEX Robotics World Championship once

SKILLS

Languages Rust, C++, C, Python, Java, HTML/CSS/JavaScript

Robotics ROS 2 (Humble, Jazzy), Isaac ROS, NVIDIA Jetson, Gazebo, SLAM/AMCL/EKF, YOLOv8, RealSense

Tools Linux administration, Docker, Git/GitHub, Django, ffmpeg, Vim, Fusion 360

ACTIVITIES

Theatre Tech, The Grayson School, 2022–2025 (sound board)